

006_REQ — Plant calendar

006_REQ version 1. SCR-006 「稼働カレンダー」 (Working calendar).

Field	Value
System / group	ProductionManagementAI / Master data
Work item	WI-010
Date / author	2026-10-01 / Agent
State	Submitted for requirements review; no implementation authorization
Plan	Revision 1 , explicitly approved
Dependency	WI-009 merged by PR #34 as 42e8932; existing plant clock and Product master

1. Objective and scope

Maintain a shared plant working calendar with line-specific date exceptions. Show working days, holidays, shutdowns and exceptional working days on desktop and mobile. Explain the effective rule and allow dated capacity lookup using current line/product configuration. Use the configured plant timezone (currently Asia/Tokyo) for all dates and for the server's definition of today.

This phase is calendar master and lookup only. Existing due-date validation, order-start rules, dashboard due-soon windows and historical order data remain unchanged. The source PDF's working-day date arithmetic is explicitly deferred by DEC-004. No automatic schedule, allocation, overload control, load chart, BOM, shift/time-of-day model, holiday-provider integration or live deployment.

2. Actors and use cases

Actor	Use case	Result
Admin / Operator	UC-019: view the effective calendar	Browse a month, select plant/line scope and see dates, working state and rule source
Admin / Operator	UC-020: maintain weekly patterns and date exceptions	Save confirmed date-bounded rules, preserve past dates and reject stale mutations
Admin / Operator	UC-021: inspect dated line/product capacity	See effective hours and descriptive quantity in the actual product unit for today/future

Both roles may read and write; every calendar API operation independently checks authorization. Other authenticated roles receive no calendar access by hiding/showing controls. Existing order/dashboard roles are unchanged.

3. Requirements and acceptance criteria

ID	Requirement	Successful path	Failure / exception path
REQ-070	View effective monthly calendar	Desktop/mobile show plant-local dates, working/nonworking state, reason where present and winning source; month/scope selection has explicit applied state	Invalid month/scope rejected; failed reads show error, not invented working days; unavailable earlier configuration identified explicitly
REQ-071	Maintain effective weekly patterns	Initial pattern Monday–Friday working, Saturday/Sunday nonworking; new pattern effective today/future; earlier effective versions retained for past lookup	Past effective changes rejected server-side, including crossing plant midnight; concurrent stale save cannot overwrite or partially apply
REQ-072	Maintain manual date exceptions	Plant or line exception can close a day or make it working, optionally replacing that day's hours; removing an editable exception restores documented lower-priority fallback	Past edits/removal, invalid date/hours or conflicting duplicate exception rejected; Cancel/Escape writes nothing; no implicit public-holiday import
REQ-073	Resolve effective hours and dated capacity	Line-date exception outranks plant-date exception, then weekly pattern; working day without explicit hours uses current WI-009 line hours; nonworking day has zero hours; usable current line/product timing gives descriptive unit-bearing capacity for today/future	Retired/nonexistent/incompatible/stale-unit configuration not presented as usable capacity; past capacity unavailable, not fabricated; quantities across products or units never added
REQ-074	Preserve secure, concurrent and accessible behavior	Japanese UI with distinct centralized icon; API/UI Admin/Operator checks; keyboard/month navigation, centered confirmation, focus return and mobile/200% zoom usability	Unauthorized operations denied; invalid data preserves inputs; conflicts require reconciliation; unknown write outcome does not trigger blind replay; no secrets/raw business values in telemetry
REQ-075	Preserve existing order and dashboard date semantics	Existing calendar-day due-date checks, due-soon window and permitted status transitions continue; calendar state only informs this feature's lookup	A nonworking date does not newly reject an order save/start; no historical order mutation, deadline shift, working-day reinterpretation or inferred scheduling

4. Confirmed business rules

[DEC-001–007](#) record explicit user feature/plan choices and approvals of the proposed business rules.

1. Read/write roles: Admin and Operator. Shared plant calendar plus line overrides.
2. Initial weekdays: Monday–Friday working; Saturday/Sunday nonworking. Editors may configure the weekly working pattern subject to effective-date protection.

3. A working day uses WI-009's current hours/day for the selected line unless a winning date exception explicitly supplies hours. Do not invent a plant-wide eight-hour baseline or overwrite the line's existing master value.
4. Explicit working-day hours: greater than zero, at most 24 hours, at most three fractional digits. Closed day has zero effective hours. Zero working hours, negative/overprecision/out-of-range hours and contradictory closed-day hours are invalid; clearing a working exception's hours restores line-hour fallback.
5. Precedence: line-date exception → plant-date exception → effective weekly pattern. A line-specific working exception can reopen a plant nonworking day. A line-specific closure can close a plant working day. Removing a date exception restores its lower-priority rule; it does not create a hidden replacement.
6. Holidays/shutdowns entered manually. No automatic Japanese holiday feed and no implied national-holiday rule. A holiday affects only its explicitly configured date(s); a UI may help enter dates without altering resolution semantics.
7. Calendar edits and exception removal for dates before server plant-local today are rejected. Weekly changes become effective today or a future date; previous effective versions remain available. Client timezone never changes this cutoff.
8. Dated capacity lookup is only for today/future using current valid line/product configuration, including current unit generation. Calendar past display is not a historical capacity, actual production result or utilization record.
9. Calendar rules do not gate order starts or alter order/dashboard date semantics.

5. Review proposals for initialization and lookup presentation

These complete the acceptance boundary for this requirements review. They are explicit proposals rather than additional user business answers; dependent design waits for review of this document.

ID	Proposal	Reason / observable result
RP-01	Initialize the first weekly rule at the plant-local activation date; dates before the first recorded rule show configuration unavailable, unless a retained applicable rule exists	Avoid fabricating a calendar the application never stored; before activation, historical default is not a claim the plant operated
RP-02	At most one current exception for each scope/date; replacing/removing today's or future exception preserves prior stored revisions; scheduled weekly patterns may be corrected/withdrawn before their effective date while retaining prior records	Prevent ambiguous resolution; removal restores fallback, without destroying the retained definition of past dates
RP-03	Calendar scope view without a selected line shows working state and explicit exception hours, or line-dependent hours; it never displays an invented universal hour count	Default hours exist on WI-009 line masters, not on the weekly weekday mask
RP-04	For usable line/product pair, descriptive capacity = effective hours × 60 ÷ minutes per product unit; presentation rounds down to whole units for discrete units, or at most three decimals for kg/m	Stay consistent with existing order quantity precision; show product unit and state that current capacity is not reserved or historical

ID	Proposal	Reason / observable result
RP-05	Retired lines remain identifiable in retained calendar entries but are not candidates for new line exceptions or usable capacity; removing an editable future retired-line exception is allowed	Preserve references while preventing new operational use; past-date protection still applies

DB/API/FN designs will specify technical length/date-range/paging limits, exact numeric transport, concurrency tokens, boundary transactions and retention representation. They must preserve these reviewed business outcomes; no contract or implementation is claimed by this document. Generic change-history UI/export is outside this scope; retained revisions support calendar resolution protection.

6. Information and interaction boundaries

SCR-006 「稼働カレンダー」 (Working calendar), proposed route `/plant-calendar`. Display calendar date, scope, effective working state/hours, source, reason and appropriate editability. Weekly configuration and date-exception editors are part of this feature. Capacity lookup includes line/product/date and product unit; no sum across products and no assumed physical device capture.

Monthly navigation must be keyboard usable; distinguish working/closed/unavailable states with text as well as color. Explicit loading, empty, validation, forbidden, read error, conflict, pending and unknown-outcome states preserve honest data. Past dates readable where configured, with edit controls disabled and API rejection independent from the UI. Reconcile drafts before reload or leaving dirty forms.

Use existing cookie authentication, same-origin JSON mutations and server-side validation. Keep telemetry dimensions bounded; no raw dates, search terms, names, reasons, quantities, user credentials or exception details as metric labels.

7. Traceability and verification prerequisites

Trace REQ-070–075 into 006_BD/DB/main DD/API/FN/SPD and the future test plan. UC-019–021 are reserved for this work item. No existing screen behavior change requires rewriting approved completed documents. Step 10 impact addendum may be not applicable if all capacity/calendar interactions remain inside SCR-006.

Planned checks: weekday and leap/month/year boundaries; plant/browser timezone and midnight; precedence and reopening/removal; explicit hours/precision; initial unavailable dates; version transitions and protected past dates; stale saves and concurrent mutation; no invented current/past capacity; unit-change ABA and retired references; API read/write roles; keyboard/dialog/mobile/zoom; existing order and dashboard regression. None has run for WI-010; the calendar is not implemented.

New persistence must be additive and initialized with explicit activation policy, without rewriting old orders, line hours, product units or approved designs. Migration/rollback and owner permissions are defined later, with isolated rehearsal only after a separately approved implementation plan. No live deployment.

This source and its English/Japanese PDFs are the current sequential review package. Requirements review approval authorizes the next step within revision 1; it does not authorize application implementation or remote delivery.